

Lab W2D3: Containerise – Evidence Report

1. Docker Image Build

```
(base) n00n1@Nadia:~/serving-stack$ nano Dockerfile
(base) n00n1@Nadia:~/serving-stack$ docker build -t n00n1/aidc-serving:cpu-v1 .
[+] Building 147.5s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                 0.0s
=> => transferring dockerfile: 489B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 0.1s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 101B                                     0.0s
=> CACHED [1/6] FROM docker.io/library/python:3.11-slim@sha256:9c900dea9e8fb7e16277c179b555cc72d29a352dbc33cff48 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:9c900dea9e8fb7e16277c179b555cc72d29a352dbc33cff48ad5a041 0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 353B                                     0.0s
=> [2/6] RUN useradd --create-home app && mkdir -p /home/app/.cache/huggingface && chown -R app:app /hom 0.2s
=> [3/6] WORKDIR /app                                              0.0s
=> [4/6] COPY app/requirements.txt .                                0.0s
=> [5/6] RUN pip install --no-cache-dir --index-url https://download.pytorch.org/whl/cpu --extra-i 101.7s
=> [6/6] COPY app/ .                                              0.2s
=> exporting to image                                             44.9s
=> => exporting layers                                             37.6s
=> => exporting manifest sha256:1708c0c774eb57aece071a3dee94f36126ea6a84d76f6ac76d857ea7ffa00caf 0.0s
=> => exporting config sha256:3d499913c2201cffe9e4d87754a035e0fde09bf3618439f2813d3ad09a480233 0.0s
=> => exporting attestation manifest sha256:3601976c4589d1879aff52db415c06f6421310f68c650dbe653fdde4e7436fd9 0.0s
=> => exporting manifest list sha256:a3af1442c1bdf076afef422969f3c9b5f46934153c9d90c478b6c58800a17ba7 0.0s
=> => naming to docker.io/n00n1/aidc-serving:cpu-v1              0.0s
=> => unpacking to docker.io/n00n1/aidc-serving:cpu-v1          7.2s
```

The Docker image was built successfully using the optimized Dockerfile.

The image was created with:

`docker build -t n00n1/aidc-serving:cpu-v1 .`

The Dockerfile uses:

- Python 3.11 slim base image
- CPU PyTorch packages
- Layer caching optimization
- No pip cache storage

Result:

Docker image created successfully:

`n00n1/aidc-serving:cpu-v1`

2. Model Cache Volume and Container Deployment

```

hf-cache
(base) n00n1@Nadia:~/serving-stack$ docker run -d --name serving -p 8000:8000 \
-v hf-cache:/home/app/.cache/huggingface \
n00n1/aidc-serving:cpu-v1
9d12e951385b287a94f5c9f1d9010660f5c1cef0c7567b7a4ae6ee74dcc62c07
(base) n00n1@Nadia:~/serving-stack$ docker logs -f serving
INFO:      Started server process [1]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)

```

A Docker volume was created to store the Hugging Face model cache outside the Docker image.

The volume was created using:

```
docker volume create hf-cache
```

The container was started with the mounted cache volume:

```

docker run -d --name serving -p 8000:8000 \
-v hf-cache:/home/app/.cache/huggingface \
n00n1/aidc-serving:cpu-v1

```

This ensures that model weights remain outside the image and can be reused between container runs.

3. Container Running Successfully

```

n00n1/aidc-serving:cpu-v1
9d12e951385b287a94f5c9f1d9010660f5c1cef0c7567b7a4ae6ee74dcc62c07
(base) n00n1@Nadia:~/serving-stack$ docker logs -f serving
INFO:      Started server process [1]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
^C(base) n00n1@Nadia:~/serving-stack$ curl -s http://localhost:8000/v1/chat/completions \
-H "Content-Type: application/json" \

```

The container started successfully and the FastAPI service became available.

Container logs confirmed:

Application startup complete.

Uvicorn running on http://0.0.0.0:8000

The service is now running inside the Docker container.

4. API Health Check

```
(base) n00n1@Nadia:~$ curl http://localhost:8000/health
{"status":"ok","model":"Qwen/Qwen2.5-0.5B-Instruct"}(base)
```

The health endpoint responded successfully, confirming that the model service is running correctly.

5. Chat Completion API Test

```
INFO: OpenAI LLM running on http://0.0.0.0:8000 (Press ENTER to quit)
^C(base) n00n1@Nadia:~/serving-stack$ curl -s http://localhost:8000/v1/chat/completions \
-H "Content-Type: application/json" \
-d '{"model":"Qwen/Qwen2.5-0.5B-Instruct","messages":[{"role":"user","content":"Say hi."}], "max_tokens":16}'
{"id":"chatcmpl-520d05d12af04ffd90faae09f89585f8","object":"chat.completion","created":1787657086,"model":"Qwen/Qwen2.5-0.5B-Instruct","choices":[{"index":0,"message":{"role":"assistant","content":"Hello! How can I assist you today?"},"finish_reason":"stop"}],"usage":{"prompt_tokens":32,"completion_tokens":10,"total_tokens":42}}(base) n00n1@Nadia:~/serving-stack$ docker images | grep aidc-serving
n00n1/aidc-serving:cpu-v1                                a3af1442c1bd          1.61GB          336MB          U
```

```
curl -s http://localhost:8000/v1/chat/completions \
```

The OpenAI-compatible API endpoint was tested successfully.

6. Docker Image Size Comparison

```
tack$ docker images | grep aidc-serving
n00n1/aidc-serving:cpu-v1                                a3af1442c1bd          1.61GB          336MB          U

=> => unpacking to docker.io/library/aidc-serving:naive
(base) n00n1@Nadia:~/serving-stack$ docker images | grep aidc-serving
aidc-serving:naive                                       631f89019983          16.5GB          6.6GB
n00n1/aidc-serving:cpu-v1                                a3af1442c1bd          1.61GB          336MB          U
(base) n00n1@Nadia:~/serving-stack$
```

The optimized Docker image was compared with the naive image.

Image	Size
Naive build	16.5 GB
Slim build	1.61 GB

The optimization reduced the image size significantly by using:

- Slim base image
- No pip cache
- Optimized Docker layers

7. Docker Hub Push

```
Login Succeeded
(base) n00n1@Nadia:~/serving-stack$ docker tag n00n1/aidc-serving:cpu-v1 nadiax/aidc-serving:cpu-v1
(base) n00n1@Nadia:~/serving-stack$ docker push nadiax/aidc-serving:cpu-v1
The push refers to repository [docker.io/nadiax/aidc-serving]
b6bee9975076: Pushed
44136fa355b3: Pushed
7ce8f1a1db8f: Mounted from library/python
26c307b5e35a: Mounted from library/python
daff6f8ae2ef: Mounted from library/python
605d2cb1fe61: Mounted from library/python
93da54a25f07: Pushed
f343215cce58: Pushed
99103d961379: Pushed
2c1d0db6972d: Pushed
3a4d8aa69147: Pushed
cpu-v1: digest: sha256:a3af1442c1bdf076afef422969f3c9b5f46934153c9d90c478b6c58800a17ba7 size: 856
(base) n00n1@Nadia:~/serving-stack$
```

The Docker image was successfully pushed to Docker Hub.

8. Final Verification

```
(base) n00n1@Nadia:~/serving-stack$ python verify.py
model: Qwen/Qwen2.5-0.5B-Instruct
completion content: 'Hello!'
usage: {'prompt_tokens': 35, 'completion_tokens': 3, 'total_tokens': 38}
streaming: not implemented (optional this week)
GREEN CHECK: PASS
```